

ZERO-TRUST IMMUNE SYSTEM FOR HOSPITAL NETWORKS

Panacea

Command Center — Functionality & Working Document

An autonomous security agent that reads medical-device manuals, reasons about their real network needs, and enforces least-privilege firewall policies automatically — with every decision streamed live and permanently audited.

Version	v2
Date	July 05, 2026
Component	Command Center (Operator Dashboard)
Status	Live · deployed on Vercel

1 · Executive Summary

Medical devices are notoriously hard to secure: they run legacy firmware, expose network services described only in dense PDF manuals, and can't be patched like normal computers. Panacea treats the network as the control point. An operator drops a device's manual into the Command Center; an AI agent reads it, works out which ports the device legitimately needs, checks for known vulnerabilities, and enforces a zero-trust firewall policy that isolates the device — closing everything an attacker could use. Every step of the agent's reasoning is shown live, and every decision is recorded for compliance.

This document describes the **Command Center** — the operator-facing dashboard — what is built and working today, how the system works end-to-end, and the roadmap ahead.

2 · How It Works (End-to-End)

- Ingest.** An operator drags a device manual (PDF) into the dashboard. The full text is extracted in the browser and handed to the agent.
- Reason.** The agent (NVIDIA Nemotron on Vultr) extracts the device's network profile, cross-checks known CVEs, and deliberates over a zero-trust policy — streamed live, token-by-token, into the reasoning terminal.
- Decide.** It produces *Contract B*: a precise firewall policy (ALLOW/DENY per port), a confidence score, any flagged CVE, and a plain-English incident memo.
- Enforce.** The policy is applied at the network layer (Vultr firewall) to isolate the device to only its legitimate traffic.
- Audit.** Every decision is written to an append-only audit trail — “every autonomous action is auditable.”
- Override.** Operators can retract any active policy instantly from the dashboard.

3 · What's Built & Working — Command Center

The operator dashboard is live, deployed, and integrated with the backend over a secure tunnel.

Working capabilities:

- **Live Agent Reasoning Stream.** A real-time WebSocket terminal that shows the agent's pipeline as it happens. Raw token-by-token output is summarized into clean, Cursor-style activity rows — each milestone (extraction, CVE check, decision, enforcement) is one line you can expand; the model's chain-of-thought collapses into a single “Reasoning” block. A “Steps / Raw” toggle is available.
- **Expandable Error Log.** Any failure — backend errors, dropped connections, extraction warnings — appears as an expandable red/amber row in the terminal instead of being hidden, so operators always see the true state.
- **PDF Manual Ingestion.** Drag-and-drop upload with robust in-browser text extraction (pdf.js) that reliably reads real manufacturer PDFs, then sends a curated, relevant slice to the agent for analysis.
- **Incident Memos (Contract B).** Live decision cards showing the enforced firewall rules, a confidence gauge, CVE flags, and the agent's incident summary — plus one-click “Explain this decision”, which returns a plain-English compliance justification from the agent.
- **Decision Audit Trail.** A real, append-only history of every agent decision — timestamp, device, confidence, flagged CVE, enforced rules, and policy drift — pulled directly from the backend.
- **Device Fleet & Telemetry.** A monitored-device table with firewall status and ports, a live device-heartbeat visualization, and a network-activity chart driven by real enforcement events.
- **Human Override.** A single control to retract an active policy, mapped to the backend's policy-retraction endpoint.
- **Modern UI & Delivery.** A minimalist, dark “Composio-style” interface with tabbed navigation (Overview · Live Agent · Ingest · Devices · Audit), deployed and previewable on Vercel.

4 · Backend & Agent Capabilities

The Command Center is powered by a Vultr-native agent backend. Key capabilities the dashboard surfaces:

- **7-step orchestration pipeline** — extract device profile → store & index the manual → retrieve manufacturer evidence (RAG) → check CVEs → decide policy → enforce firewall → generate incident memo.
- **Vultr-native stack** — Serverless Inference (Nemotron) for reasoning & tool use, Vector Store for grounding in manuals and a CVE knowledge base, Object Storage for sealed evidence, and firewall enforcement at the VPC layer.
- **Structured contracts** — Contract A (extracted device network profile) and Contract B (the firewall decision) give the whole system a clean, typed interface.
- **Trust & safety features** — confidence scoring derived from reasoning depth, policy-drift detection across re-scans, cryptographically sealed evidence bundles, an append-only audit log, and an “explain this decision” endpoint for compliance officers.

5 · Demo Scenario

A medical device sits on a hospital VPC with open network ports; an attacker VM on the same network can reach it. The operator ingests the device's manual in the Command Center. The agent determines the device's legitimate ports, flags risky exposure (e.g. remote administration over SSH / port 22), and enforces a zero-trust policy that closes everything the device doesn't need — cutting off the attacker's access, live on screen, with the full reasoning and the resulting audit entry visible to the audience.

6 · Technology Stack

Frontend	Next.js (App Router), React, TypeScript, Tailwind CSS, Framer Motion, Recharts, pdf.js — deployed on Vercel
Backend	FastAPI (Python), Pydantic, WebSockets
AI / Data	Vultr Serverless Inference (Nemotron), Vultr Vector Store (RAG), Vultr Object Storage, Vultr Firewall
Transport	REST + WebSocket streaming over a secure HTTPS/WSS tunnel

7 · Roadmap & Future Work

Priorities to take Panacea from a working demo to a fleet-scale product:

- **Universal manual ingestion** — smart section targeting plus OCR for scanned / image-based PDFs, so any manual format ingests cleanly regardless of layout.
- **Deeper extraction grounding** — a multi-vendor device library so port/protocol extraction is consistent across manufacturers.

- **Fleet-scale operations** — bulk ingestion and concurrent multi-device agent runs for whole-network protection in one pass.
- **Automated red-team loop** — tighter live integration of the attacker-simulation so “attack detected → policy auto-remediated” runs end-to-end automatically.
- **Enterprise readiness** — role-based operator authentication / SSO, richer historical analytics, and long-term persistence of decisions and evidence.

In short: the Command Center already delivers the core loop — ingest a real manual, watch the agent reason live, produce and enforce a zero-trust firewall policy, and audit every decision. The roadmap focuses on breadth (more manuals, more devices) and automation.